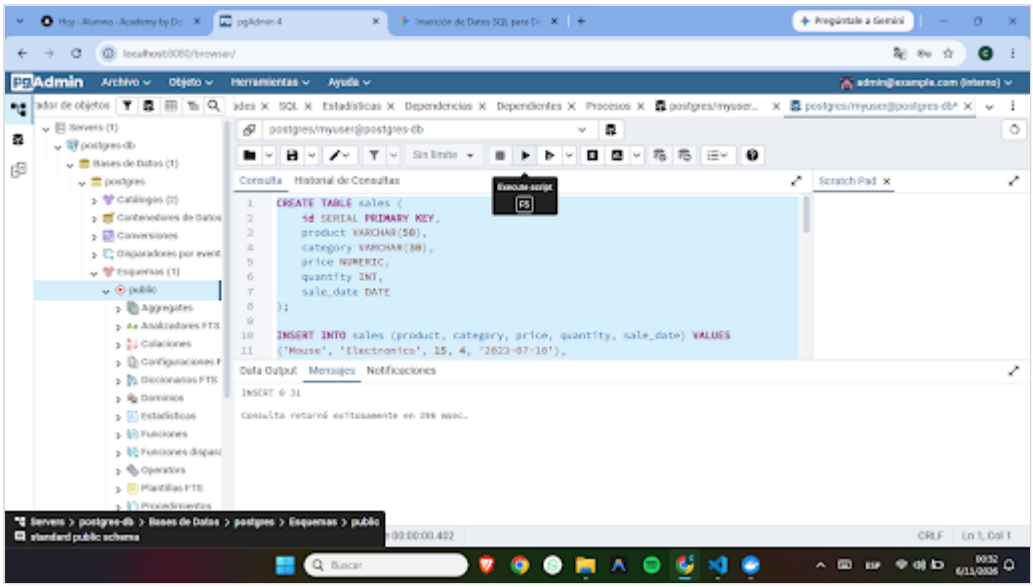
Documentación SQL: Tabla Sales (Parte 1)

0. Creación de tabla e inserción de datos

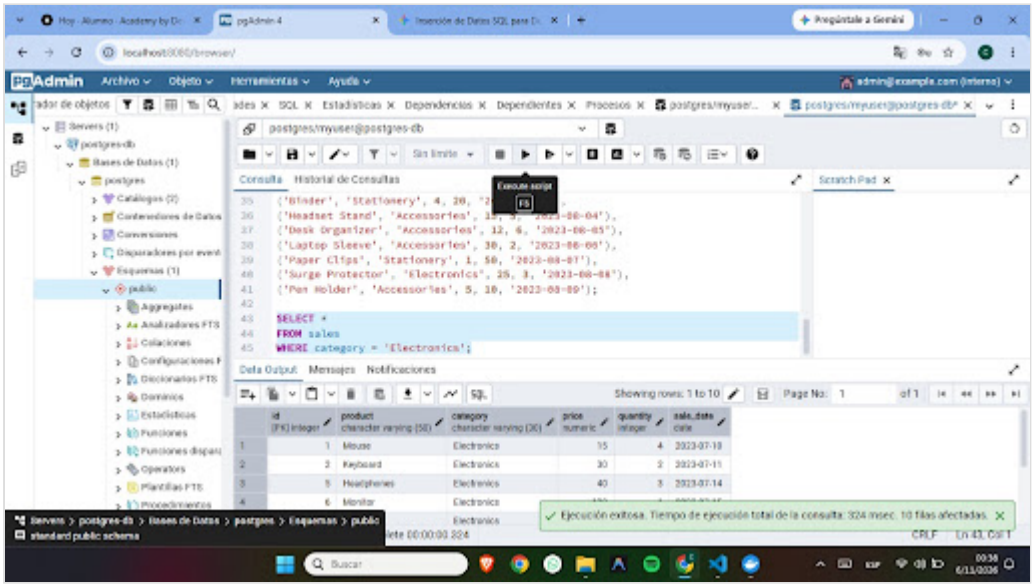
```
CREATE TABLE sales (
    id SERIAL PRIMARY KEY,
    product VARCHAR(50),
    category VARCHAR(30),
    price NUMERIC,
    quantity INT,
    sale_date DATE
);

INSERT INTO sales (product, category, price, quantity, sale_date) VALUES
('Mouse', 'Electronics', 15, 4, '2023-07-10'),
('Keyboard', 'Electronics', 30, 2, '2023-07-11'),
('Notebook', 'Stationery', 8, 10, '2023-07-12'),
('Pen', 'Stationery', 2, 25, '2023-07-13'),
('Headphones', 'Electronics', 40, 3, '2023-07-14'),
('Monitor', 'Electronics', 120, 1, '2023-07-15'),
('Desk Chair', 'Furniture', 85, 2, '2023-07-16'),
('Notebook Case', 'Accessories', 25, 3, '2023-07-17'),
('Printer Paper', 'Stationery', 6, 15, '2023-07-18'),
('USB Drive', 'Electronics', 12, 5, '2023-07-19'),
('Desk Lamp', 'Furniture', 35, 4, '2023-07-20'),
('Stapler', 'Stationery', 7, 6, '2023-07-21'),
('Calculator', 'Electronics', 20, 3, '2023-07-22'),
('Water Bottle', 'Accessories', 10, 8, '2023-07-23'),
('Laptop Stand', 'Accessories', 45, 2, '2023-07-24'),
('Office Desk', 'Furniture', 150, 1, '2023-07-25'),
('Whiteboard', 'Furniture', 60, 1, '2023-07-26'),
('Marker Set', 'Stationery', 5, 12, '2023-07-27'),
('Mouse Pad', 'Accessories', 8, 7, '2023-07-28'),
('Webcam', 'Electronics', 50, 2, '2023-07-29'),
('Smartphone', 'Electronics', 300, 1, '2023-07-30'),
('Notebook Pro', 'Electronics', 900, 1, '2023-07-31'),
('File Cabinet', 'Furniture', 120, 1, '2023-08-01'),
('Backpack', 'Accessories', 40, 3, '2023-08-02'),
('Binder', 'Stationery', 4, 20, '2023-08-03'),
('Headset Stand', 'Accessories', 15, 5, '2023-08-04'),
('Desk Organizer', 'Accessories', 12, 6, '2023-08-05'),
('Laptop Sleeve', 'Accessories', 30, 2, '2023-08-06'),
('Paper Clips', 'Stationery', 1, 50, '2023-08-07'),
('Surge Protector', 'Electronics', 25, 3, '2023-08-08'),
('Pen Holder', 'Accessories', 5, 10, '2023-08-09');
```



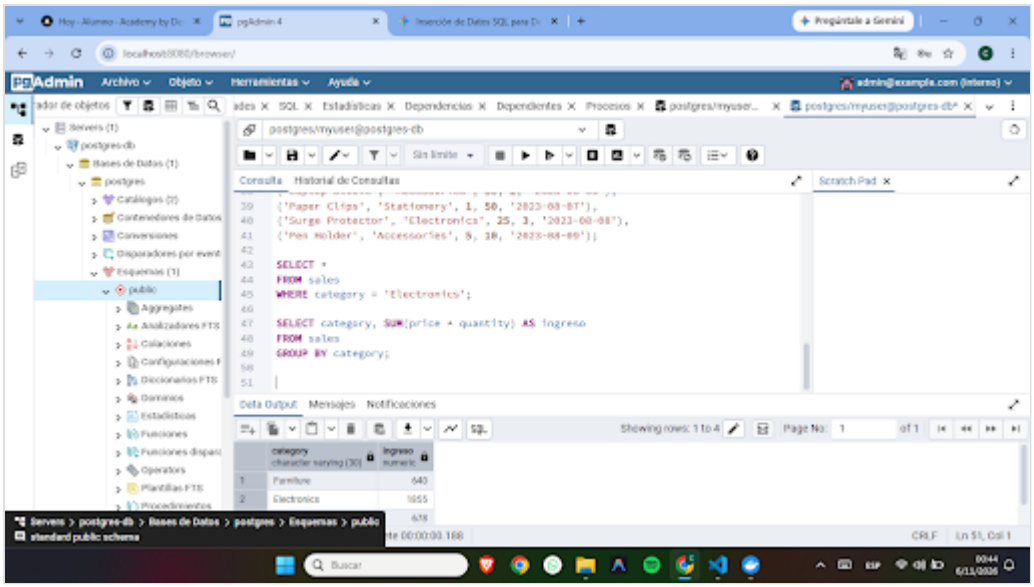
1. Las ventas de la categoría 'Electronics'

```
SELECT *
FROM sales
WHERE category = 'Electronics';
```



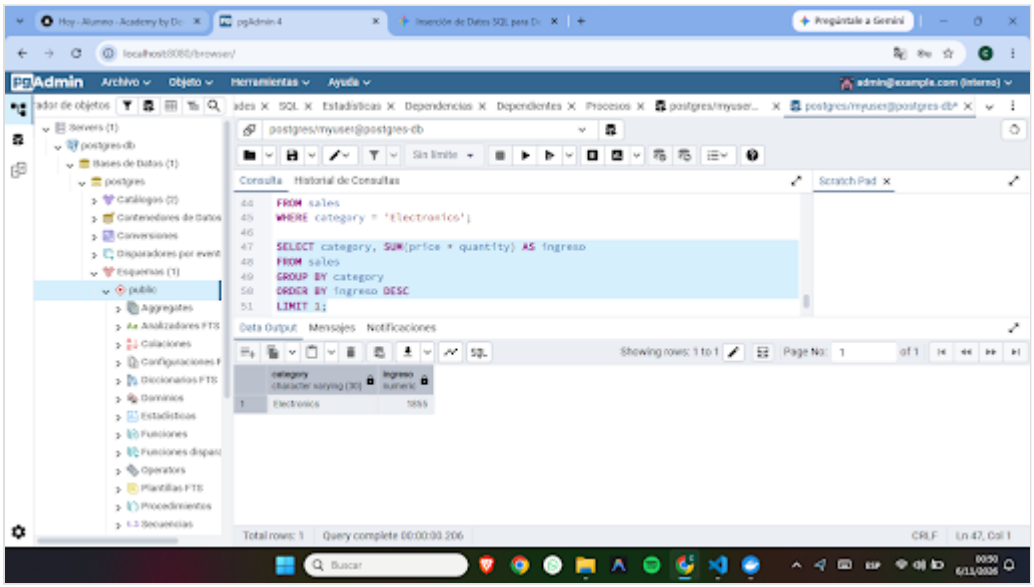
2. Total de ingresos por categoría

```
SELECT category, SUM(price * quantity) AS ingreso
FROM sales
GROUP BY category;
```



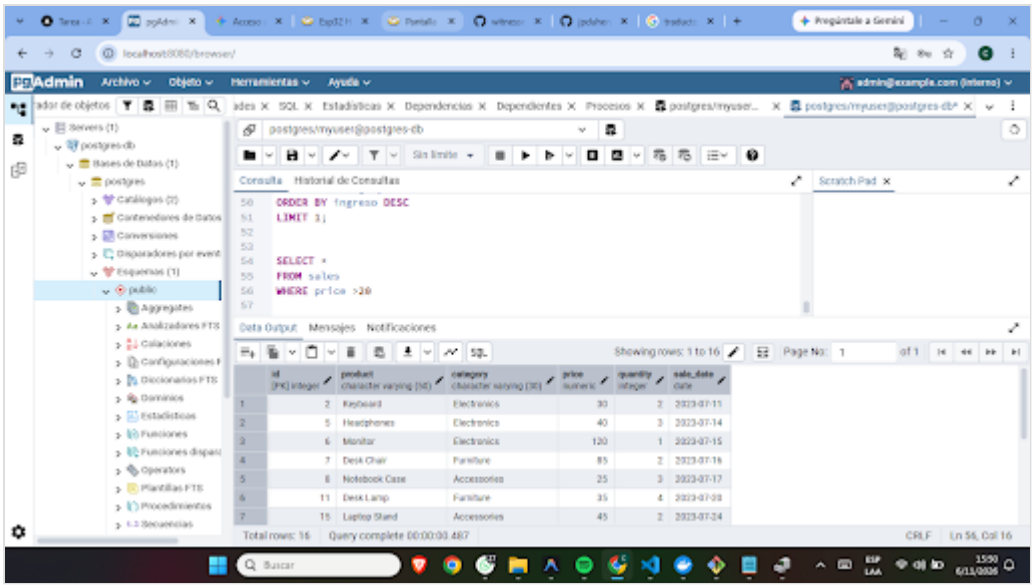
3. La categoría con mayor ingreso total

```
SELECT category, SUM(price * quantity) AS ingreso
FROM sales
GROUP BY category
ORDER BY ingreso DESC
LIMIT 1;
```



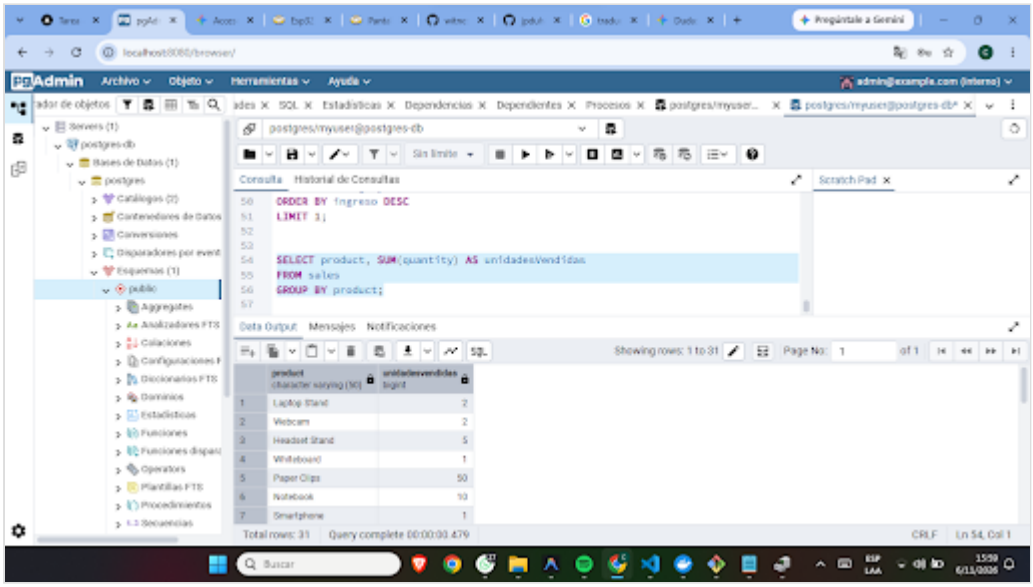
4. Ventas donde el precio es mayor a 20

```
SELECT *
FROM sales
WHERE price > 20;
```



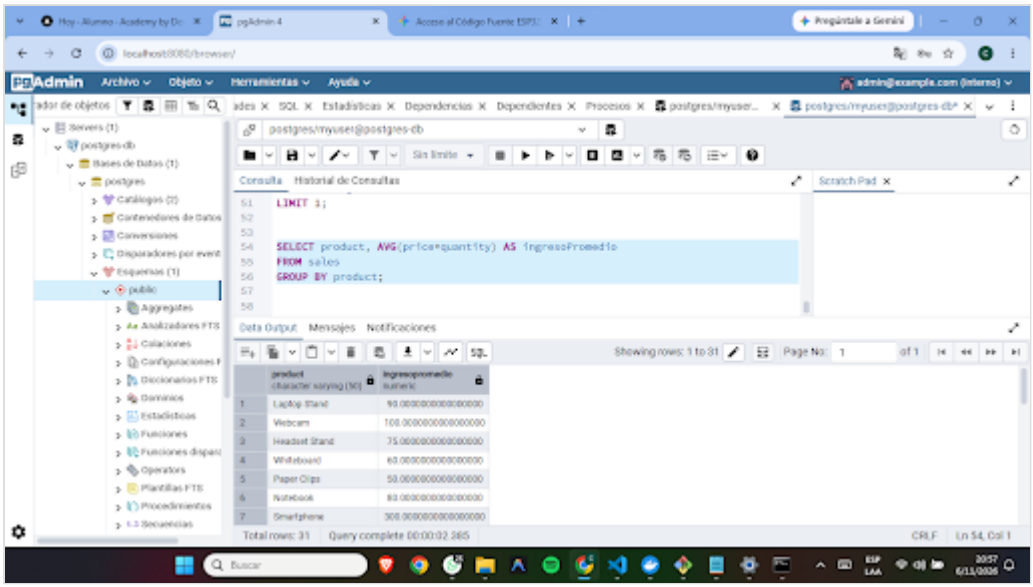
5. Total de unidades vendidas por cada producto

```
SELECT product, SUM(quantity) AS unidadesVendidas
FROM sales
GROUP BY product;
```



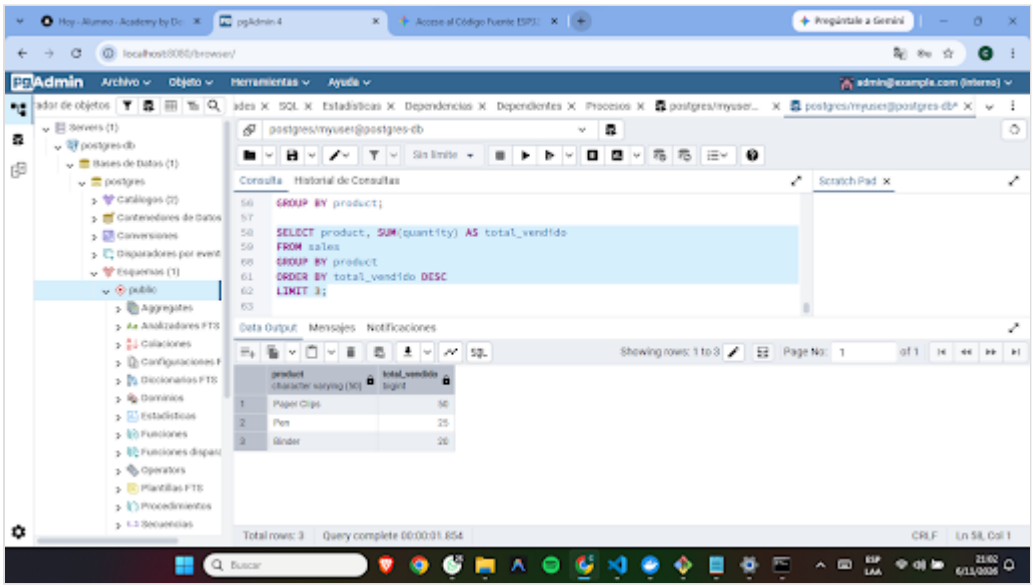
6. Ingreso promedio por venta para cada producto

```
SELECT product, AVG(price*quantity) AS ingresoPromedio
FROM sales
GROUP BY product;
```



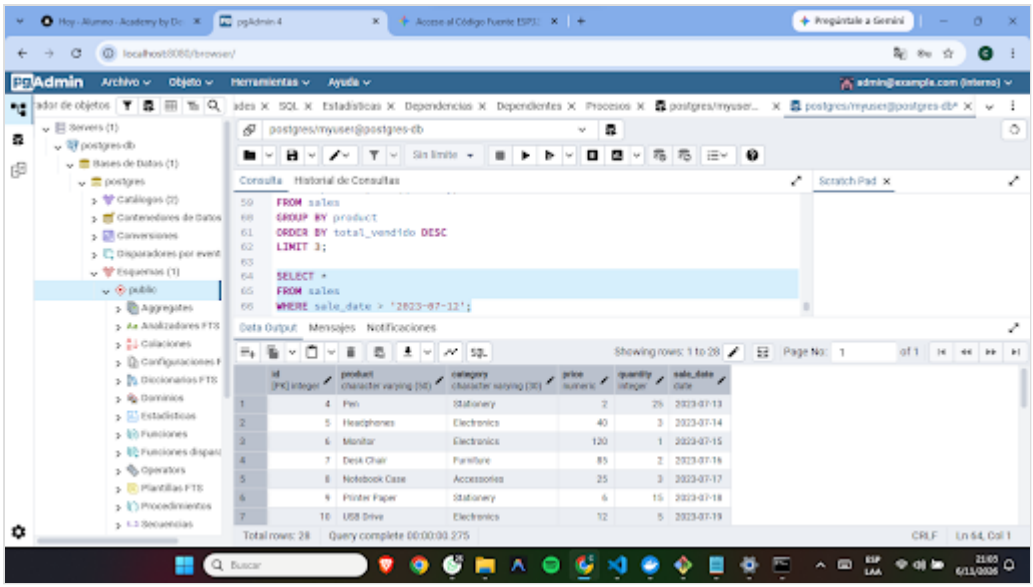
7. Top 3 productos con mayor cantidad total vendida

```
SELECT product, SUM(quantity) AS total_vendido
FROM sales
GROUP BY product
ORDER BY total_vendido DESC
LIMIT 3;
```



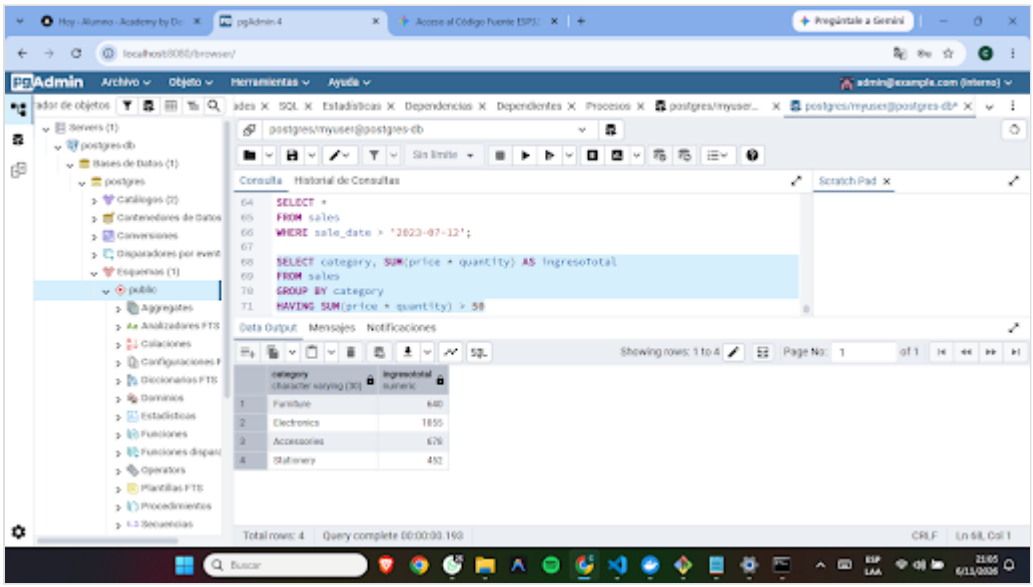
8. Ventas realizadas después del 12 de julio de 2023

```
SELECT *
FROM sales
WHERE sale_date > '2023-07-12';
```



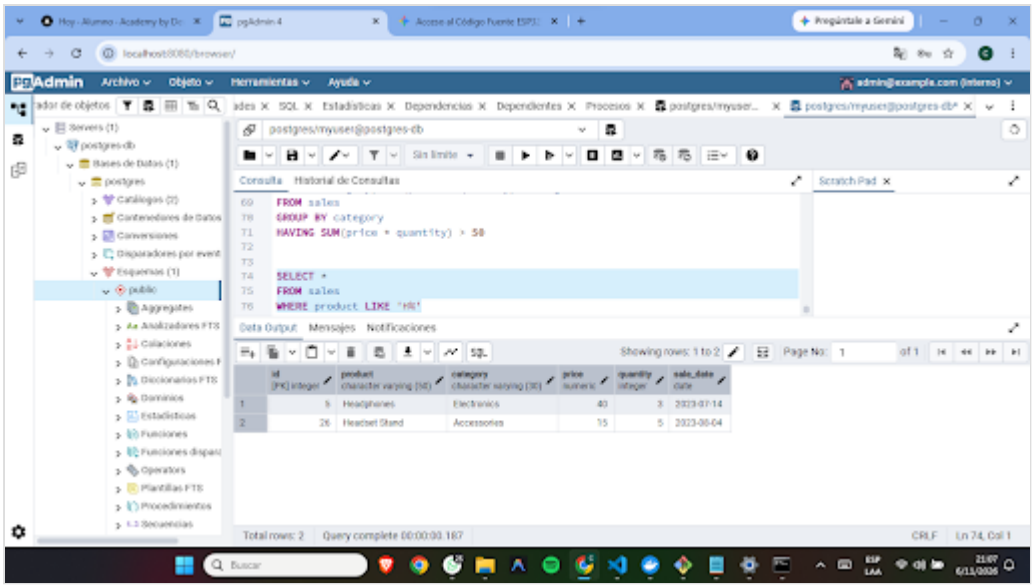
9. Ingreso total por categoría (mayor a 50)

```
SELECT category, SUM(price * quantity) AS ingresoTotal
FROM sales
GROUP BY category
HAVING SUM(price * quantity) > 50;
```



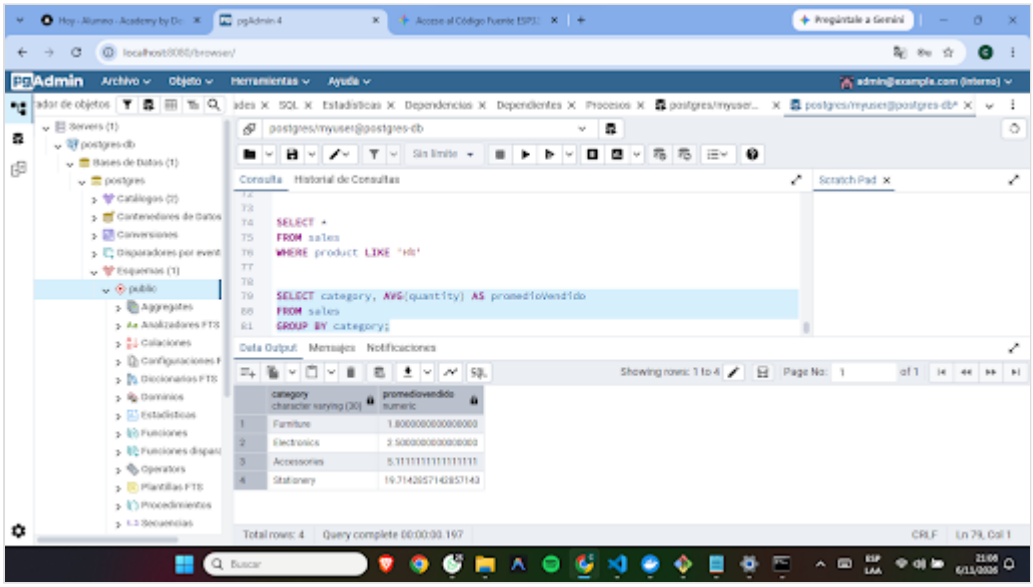
10. Productos que comienzan con la letra H

```
SELECT *
FROM sales
WHERE product LIKE 'H%';
```



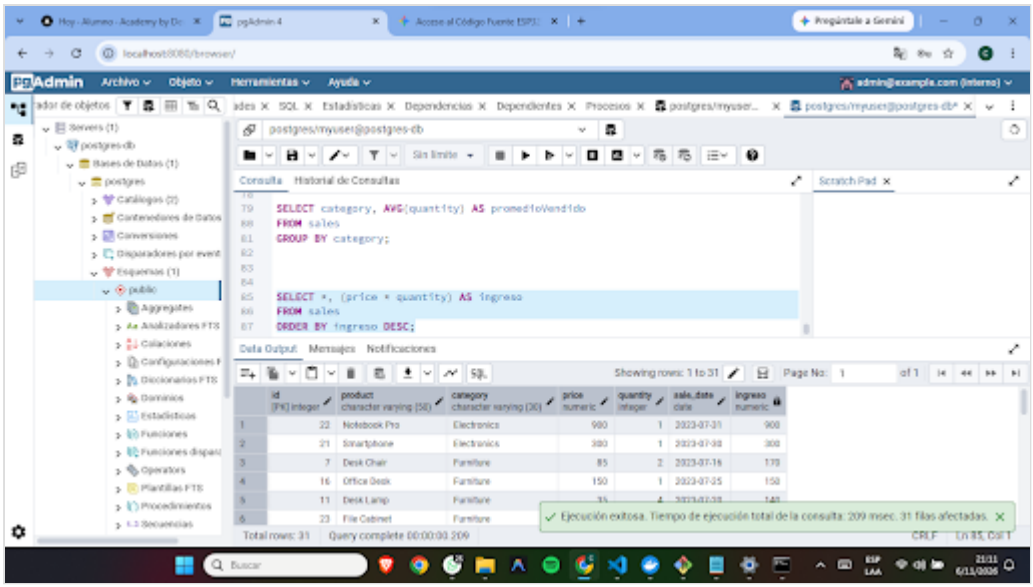
11. Promedio de cantidad vendida agrupado por categoría

```
SELECT category, AVG(quantity) AS promedioVendido
FROM sales
GROUP BY category;
```



12. Ingreso por fila ordenado de mayor a menor

```
SELECT *, (price * quantity) AS ingreso
FROM sales
ORDER BY ingreso DESC;
```



13. Ventas Electronics (cantidad > 2 y precio < 40)

```
SELECT *
FROM sales
WHERE category = 'Electronics' AND quantity > 2 AND price < 40;
```

